IMPERIAL COLLEGE LONDON

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DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

MSc and EEE PART IV: MEng and ACGI

**Corrected Copy**

# POWER SYSTEM DYNAMICS, STABILITY AND CONTROL

Friday, 17 May 10:00 am

Time allowed: 3:00 hours

**There are FOUR questions on this paper.**

**Answer ALL questions.**

*All questions carry equal marks.*

**Any special instructions for invigilators and information for candidates are on page 1.**

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## The Questions

1.

a)

i) How are transients in power networks characterised? [2]

ii) Why in extra high voltage (EHV) transmission system, do voltage transients receive the most important consideration? [3]

b)

i) What is sub synchronous resonance (SSR) in the context of power system stability? [5]

ii) How has the nature of SSR changed with the increasing presence of high-power electronics in the power network? [5]

iii) List three SSR mitigation measures. [3]

c) i) Describe the purpose of having a high gain fast acting excitation system? [3]

ii) Why is a power system stabiliser (PSS) essential when operating a large synchronous generator in a power system? [4]

[25 marks]

2.

a)

i) The rotor of a synchronous generator is of solid construction whereas the stator is made up of laminated materials. Provide technical reasons to justify such a constructional practice. [4]

ii) Besides the excitation winding, the rotor of a synchronous generator usually has additional windings – describe the purpose of such windings. [4]

iii) Why is the stator winding in a synchronous generator star connected? [3]

iii) Modern synchronous generators are designed to have a large steady state synchronous reactance (between 1.0 to 2.0 pu). However, when a direct terminal short circuit occurs, the initial currents are several times the normal load current. Explain why this is so. [5]

iv) Why does a synchronous generator offer different overload limits in the lagging and leading power factor mode of operation? [4]

Two synchronous generators of identical capacities are delivering similar amount of load to a grid network. One of them is non-salient pole (round rotor) and other one is salient pole (non-round rotor). Excitation is suddenly lost in both cases while the turbine torques continue. Which one will go out of synchronism faster? Justify your answer with reason. [1+4]

[25 marks]

3.

a)

i) Why is it difficult to analyse the operating characteristic of a synchronous machine in a stationary 3-phase (a-b-c) reference frame? [3]

ii) How does a d-q reference frame (Park reference frame) help to overcome such difficulty? [3]

iii) Despite having represented a synchronous machine in d-q reference frame, why another transformation is necessary when the machine is connected to a large power system network? [3]

iv) Derive the relationship between the quantities in two reference frames when the angle difference between the two reference frames is $\delta$-$\theta$. [5]

v) A synchronous generator supports a voltage of 15.75 kV line to line at the generator terminal bus. This bus is part of a large network which has a reference bus at another location. The power flow solution produces an angle ($\theta$) of 7 degrees for this bus voltage. The generator load angle ($\delta$) is 33 degree. Compute the direct and quadrature axis components of this terminal voltage. [5]

b)

i) Why are Type 3 and Type 4 wind turbine generators preferred over Type 1 and Type 2 in modern wind farms? [3]

ii) In a large wind farm, the wind velocity is just 1m/s more than the cut in velocity. There are 10 rows of turbine. It was observed that some turbines in the subsequent rows were not producing power. Explain clearly what could be the reason(s). [3]

[25 marks]

4.

a)

i) What is the function of a gearbox in a wind turbine generator? [2]

ii) The gear box is not necessary for some types of wind turbine generators. Explain the reason behind such an option. [2]

iii) How does one model the gear box for stability studies? [2]

iv) Why does Type 4 wind turbine usually have large number of poles? [3]

b)

i) How does stall control differ from pitch control? [3]

ii) How is passive stall control implemented? [3]

iii) Why is it necessary to have a delay and dead-band to represent pitch control? [3]

iv) Why is generator torque/power control preferred over pitch control at sub-rated wind condition? [3]

c)

How is it possible to operate a Type-3 (DFIG) machine at lower than synchronous speed (positive slip) in generator mode? [4]

[25 marks]